Screenshots

Tutorial Assignment - 2: Data Warehousing and Data Mining
Orange Data Mining - Experiments 1 to 9

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Dataset

retail_transactions (original dataset: 400 rows x 10 columns)

	TransactionID	Date	Store	CustomerID	Product	Category	Quantity	UnitPrice	TotalSales	PaymentMethod
1	T1001	2026-03-12	Store B	C014	Apples	Produce	2	1.2	2.4	Cash
2	T1001	2026-03-12	Store B	C014	Bananas	Produce	3	0.8	2.4	Cash
3	T1001	2026-03-12	Store B	C014	Juice	Beverages	1	2.6	2.6	Cash
4	T1002	2026-01-24	Store B	C078	Bread	Bakery	1	1.8	1.8	UPI
5	T1002	2026-01-24	Store B	C078	Eggs	Dairy	1	3.2	3.2	UPI
6	T1002	2026-01-24	Store B	C078	Cheese	Dairy	2	3.7	7.4	UPI
7	T1003	2026-04-25	Store A	C090	Rice	Grocery	1	5.5	5.5	Cash
8	T1003	2026-04-25	Store A	C090	Milk	Dairy	1	2.5	2.5	Cash
9	T1003	2026-04-25	Store A	C090	Bread	Bakery	1	1.8	1.8	Cash
10	T1004	2026-04-08	Store C	C078	Soap	Household	1	1.4	1.4	Card
11	T1004	2026-04-08	Store C	C078	Shampoo	Household	1	4.8	4.8	Card
12	T1005	2026-04-07	Store C	C080	Chips	Snacks	3	1.6	4.8	UPI
13	T1005	2026-04-07	Store C	C080	Chocolate	Snacks	1	2.3	2.3	UPI
14	T1005	2026-04-07	Store C	C080	Juice	Beverages	1	2.6	2.6	UPI
15	T1006	2026-02-28	Store A	C013	Apples	Produce	1	1.2	1.2	Cash
16	T1006	2026-02-28	Store A	C013	Bananas	Produce	1	0.8	0.8	Cash
17	T1006	2026-02-28	Store A	C013	Juice	Beverages	3	2.6	7.8	Cash
18	T1007	2026-02-23	Store A	C082	Rice	Grocery	1	5.5	5.5	UPI
19	T1007	2026-02-23	Store A	C082	Pasta	Grocery	2	2.1	4.2	UPI
20	T1007	2026-02-23	Store A	C082	Chips	Snacks	1	1.6	1.6	UPI
21	T1008	2026-06-13	Store B	C042	Bread	Bakery	3	1.8	5.4	UPI
22	T1008	2026-06-13	Store B	C042	Eggs	Dairy	3	3.2	9.6	UPI

Fig 0 Original dataset before any pre-processing (first 22 of 400 rows)

Experiment 1: Data Cleaning

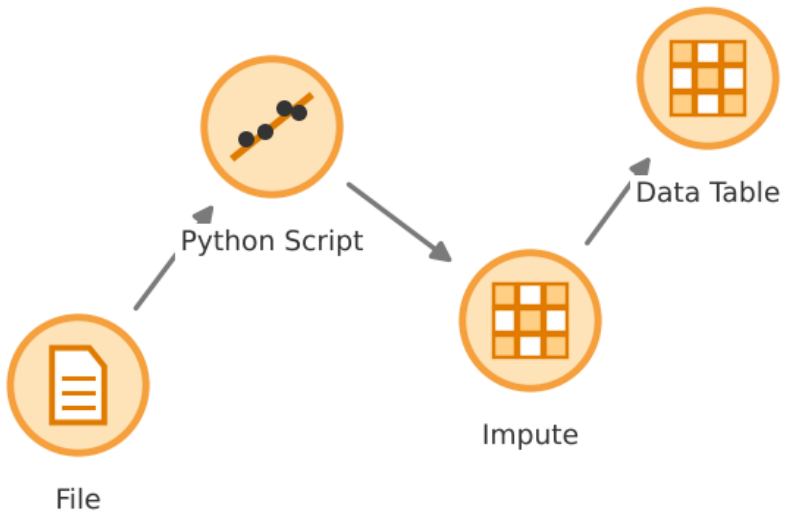


Fig 1.1 Workflow: File -> Python Script -> Impute -> Data Table

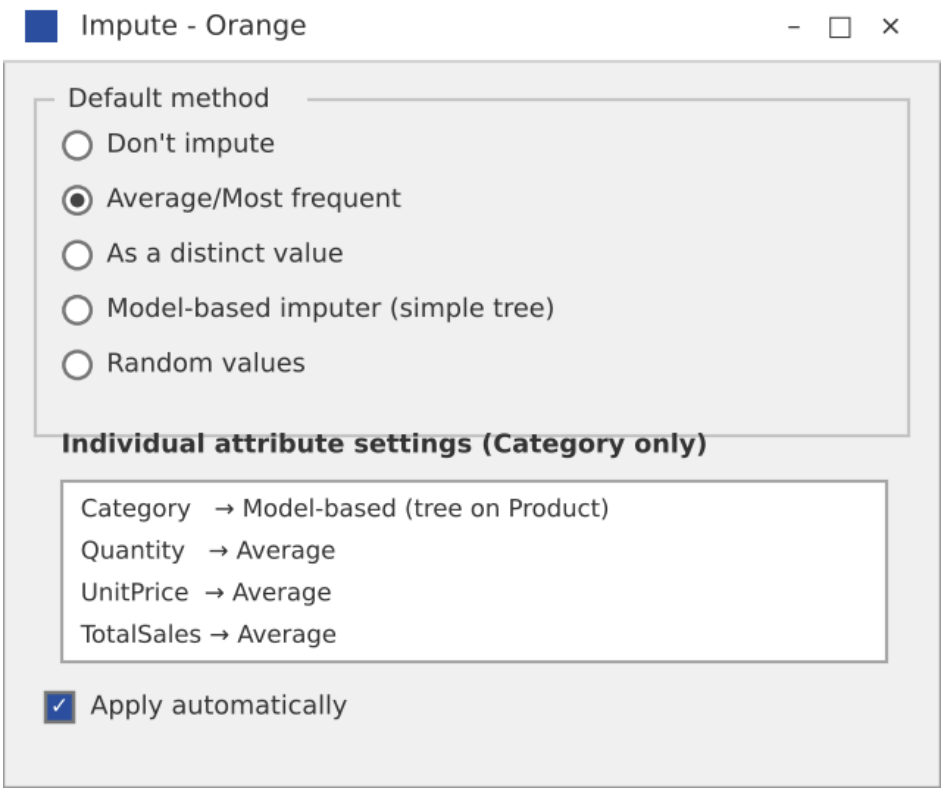


Fig 1.2 Impute widget settings

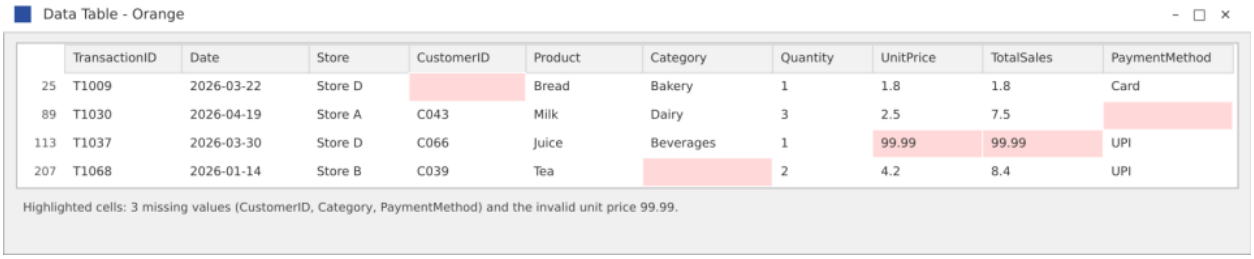


Fig 1.3 Rows with missing / invalid values before cleaning

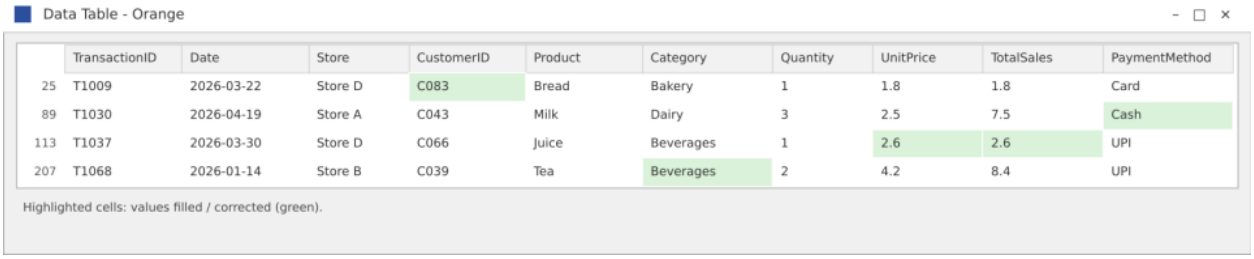


Fig 1.4 The same rows after cleaning

Experiment 2: Data Reduction

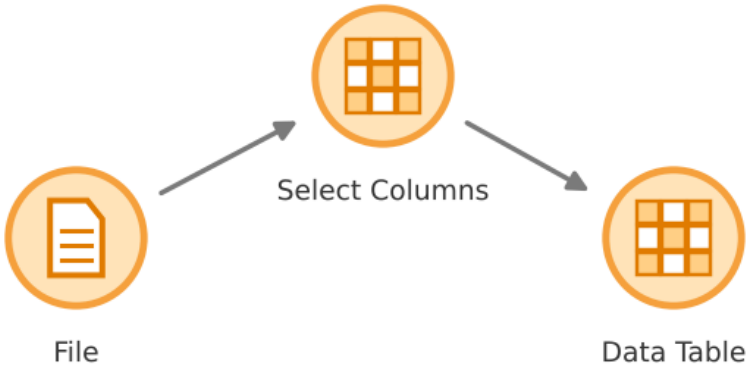


Fig 2.1 Workflow: File -> Select Columns -> Data Table

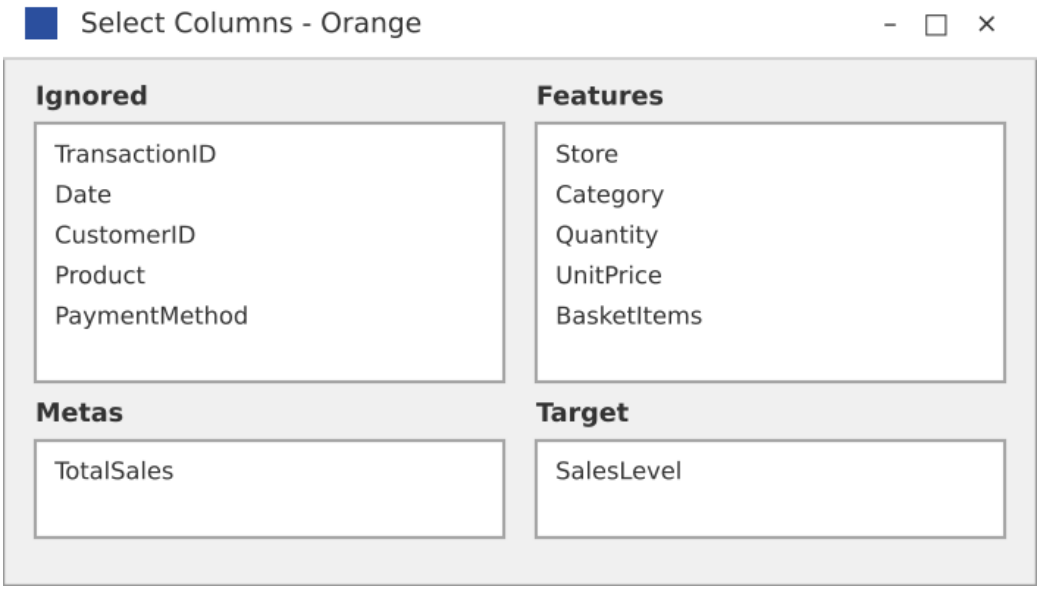
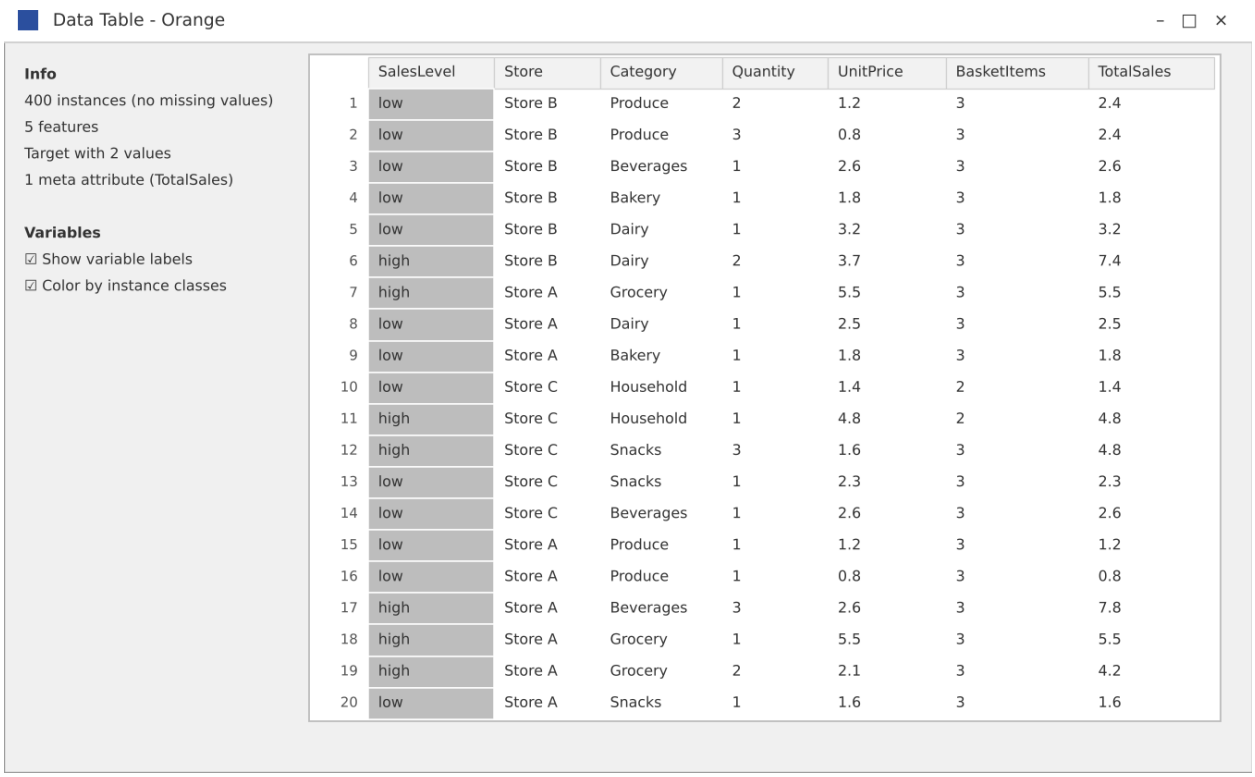


Fig 2.2 Select Columns settings



	SalesLevel	Store	Category	Quantity	UnitPrice	BasketItems	TotalSales
1	low	Store B	Produce	2	1.2	3	2.4
2	low	Store B	Produce	3	0.8	3	2.4
3	low	Store B	Beverages	1	2.6	3	2.6
4	low	Store B	Bakery	1	1.8	3	1.8
5	low	Store B	Dairy	1	3.2	3	3.2
6	high	Store B	Dairy	2	3.7	3	7.4
7	high	Store A	Grocery	1	5.5	3	5.5
8	low	Store A	Dairy	1	2.5	3	2.5
9	low	Store A	Bakery	1	1.8	3	1.8
10	low	Store C	Household	1	1.4	2	1.4
11	high	Store C	Household	1	4.8	2	4.8
12	high	Store C	Snacks	3	1.6	3	4.8
13	low	Store C	Snacks	1	2.3	3	2.3
14	low	Store C	Beverages	1	2.6	3	2.6
15	low	Store A	Produce	1	1.2	3	1.2
16	low	Store A	Produce	1	0.8	3	0.8
17	high	Store A	Beverages	3	2.6	3	7.8
18	high	Store A	Grocery	1	5.5	3	5.5
19	high	Store A	Grocery	2	2.1	3	4.2
20	low	Store A	Snacks	1	1.6	3	1.6

Fig 2.3 Data Table after column selection

Experiment 3: Data Transformation

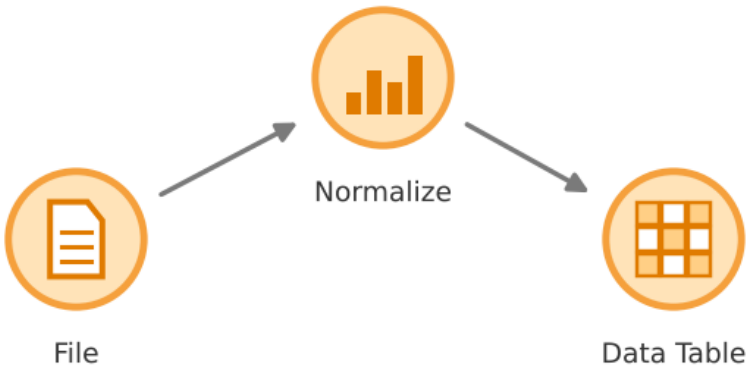


Fig 3.1 Workflow: File -> Normalize -> Data Table

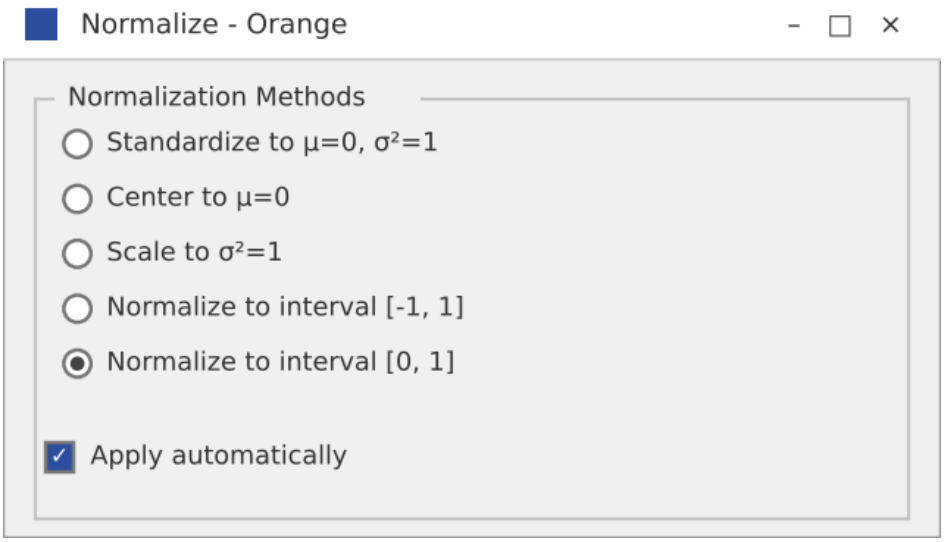


Fig 3.2 Normalize settings

Data Table - Orange

	SalesLevel	Product	Quantity	UnitPrice	TotalSales
1	low	Apples	2	1.2	2.4
2	low	Bananas	3	0.8	2.4
3	low	Juice	1	2.6	2.6
4	low	Bread	1	1.8	1.8
5	low	Eggs	1	3.2	3.2
6	high	Cheese	2	3.7	7.4
7	high	Rice	1	5.5	5.5
8	low	Milk	1	2.5	2.5
9	low	Bread	1	1.8	1.8
10	low	Soap	1	1.4	1.4
11	high	Shampoo	1	4.8	4.8
12	high	Chips	3	1.6	4.8
13	low	Chocolate	1	2.3	2.3
14	low	Juice	1	2.6	2.6

Fig 3.3 Data Table before normalization

Data Table - Orange

	SalesLevel	Product	Quantity	UnitPrice	TotalSales
1	low	Apples	0.571	0.066	0.323
2	low	Bananas	0.714	0	0.323
3	low	Juice	0.429	0.295	0.328
4	low	Bread	0.429	0.164	0.306
5	low	Eggs	0.429	0.393	0.344
6	high	Cheese	0.571	0.475	0.457
7	high	Rice	0.429	0.77	0.406
8	low	Milk	0.429	0.279	0.325
9	low	Bread	0.429	0.164	0.306
10	low	Soap	0.429	0.098	0.296
11	high	Shampoo	0.429	0.656	0.387
12	high	Chips	0.714	0.131	0.387
13	low	Chocolate	0.429	0.246	0.32
14	low	Juice	0.429	0.295	0.328

Fig 3.4 Data Table after normalization

Experiment 4: Association Rules (Apriori)

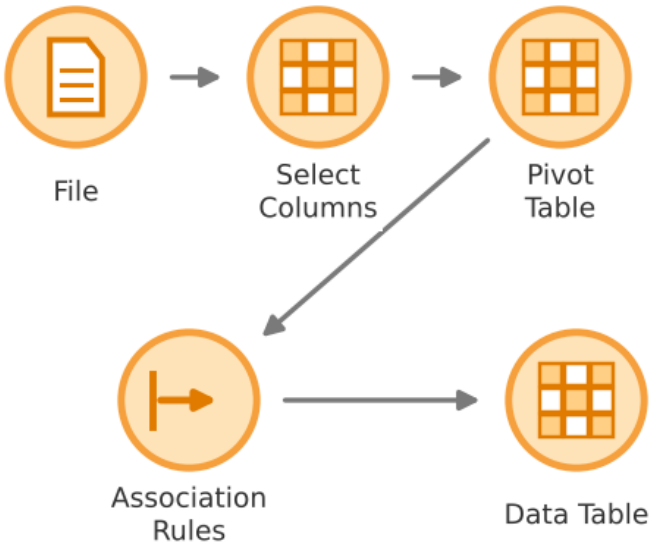


Fig 4.1 Workflow: File -> Select Columns -> Pivot Table -> Association Rules -> Data Table

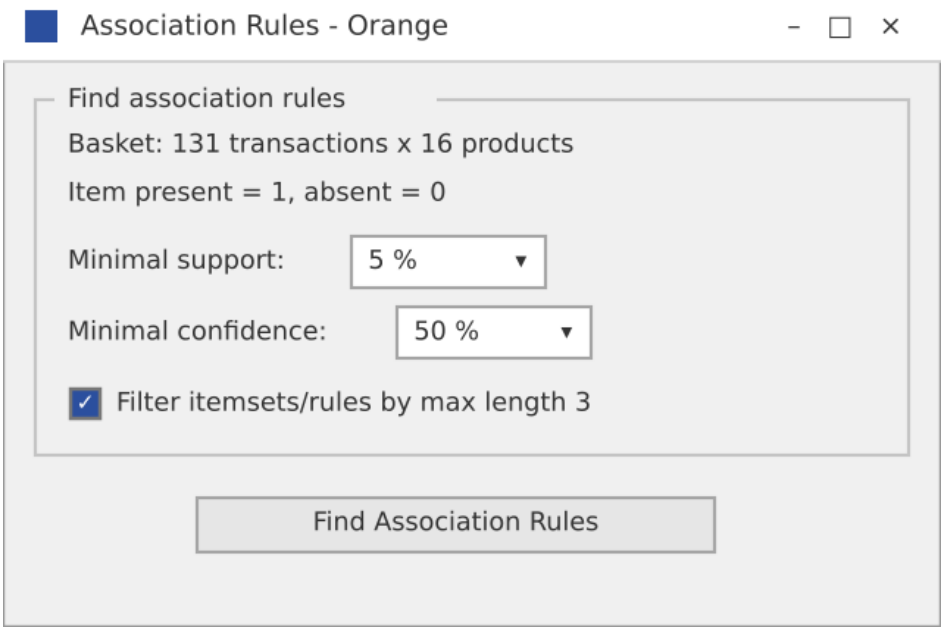
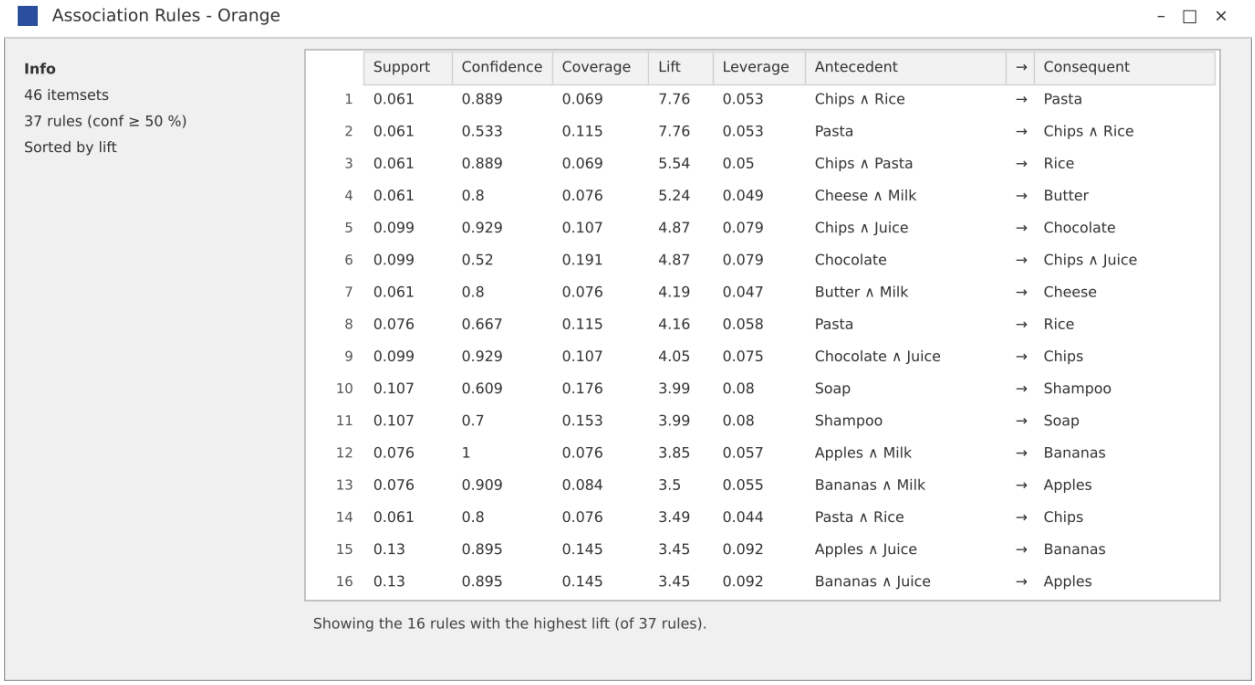


Fig 4.2 Association Rules settings



	Support	Confidence	Coverage	Lift	Leverage	Antecedent	→	Consequent
1	0.061	0.889	0.069	7.76	0.053	Chips ∧ Rice	→	Pasta
2	0.061	0.533	0.115	7.76	0.053	Pasta	→	Chips ∧ Rice
3	0.061	0.889	0.069	5.54	0.05	Chips ∧ Pasta	→	Rice
4	0.061	0.8	0.076	5.24	0.049	Cheese ∧ Milk	→	Butter
5	0.099	0.929	0.107	4.87	0.079	Chips ∧ Juice	→	Chocolate
6	0.099	0.52	0.191	4.87	0.079	Chocolate	→	Chips ∧ Juice
7	0.061	0.8	0.076	4.19	0.047	Butter ∧ Milk	→	Cheese
8	0.076	0.667	0.115	4.16	0.058	Pasta	→	Rice
9	0.099	0.929	0.107	4.05	0.075	Chocolate ∧ Juice	→	Chips
10	0.107	0.609	0.176	3.99	0.08	Soap	→	Shampoo
11	0.107	0.7	0.153	3.99	0.08	Shampoo	→	Soap
12	0.076	1	0.076	3.85	0.057	Apples ∧ Milk	→	Bananas
13	0.076	0.909	0.084	3.5	0.055	Bananas ∧ Milk	→	Apples
14	0.061	0.8	0.076	3.49	0.044	Pasta ∧ Rice	→	Chips
15	0.13	0.895	0.145	3.45	0.092	Apples ∧ Juice	→	Bananas
16	0.13	0.895	0.145	3.45	0.092	Bananas ∧ Juice	→	Apples

Fig 4.3 Association rules (support, confidence, coverage, lift, leverage)

Experiment 5: Statistical Analysis

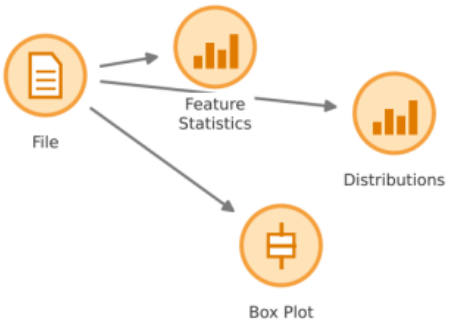


Fig 5.1 Workflow

Feature Statistics - Orange

Name	Distribution	Mean	Median	Dispersion	Min.	Max.	Missing
Quantity		1.84	1	0.585	-2	5	0
UnitPrice		2.67	2.5	0.563	0.8	6.9	0
TotalSales		4.95	3.6	0.884	-9.6	27.6	0
BasketItems		3.25	3	0.255	1	5	0
SalesLevel		Mode: low					0
Store		Mode: Store A					0
Category		Mode: Dairy					0
PaymentMethod		Mode: Cash					0

Fig 5.2 Feature Statistics

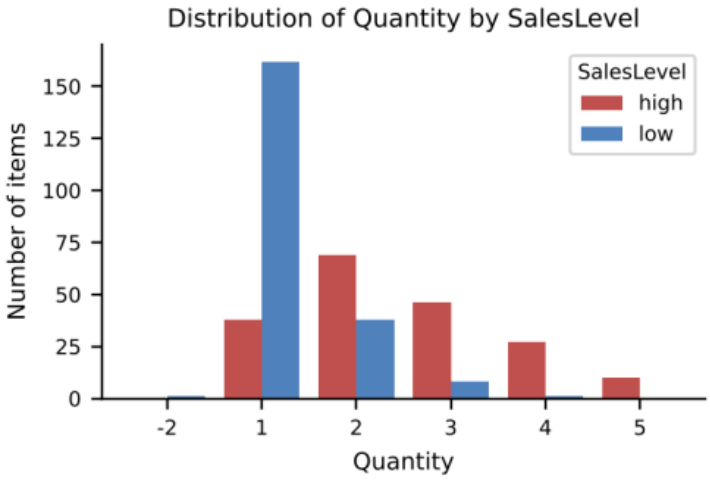


Fig 5.3 Distribution of Quantity by class

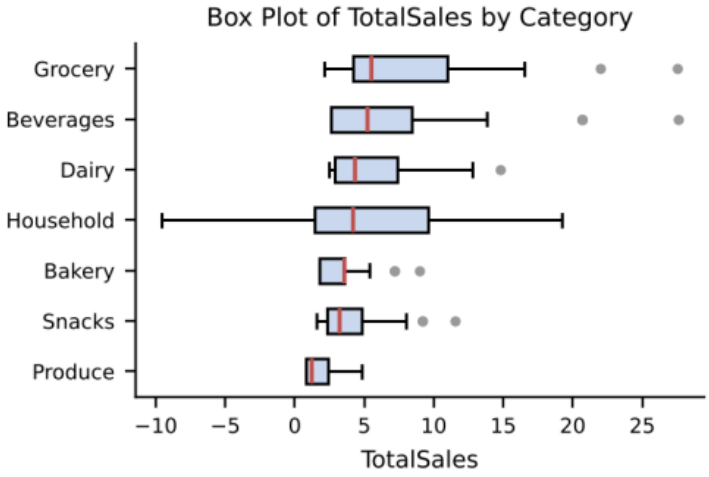


Fig 5.4 Box Plot of TotalSales by Category

Experiment 6: Regression

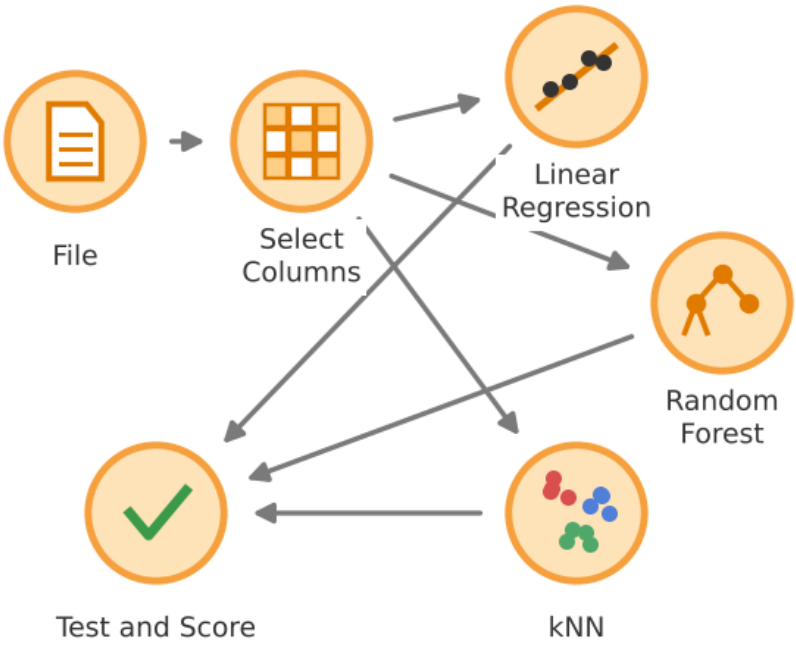


Fig 6.1 Workflow

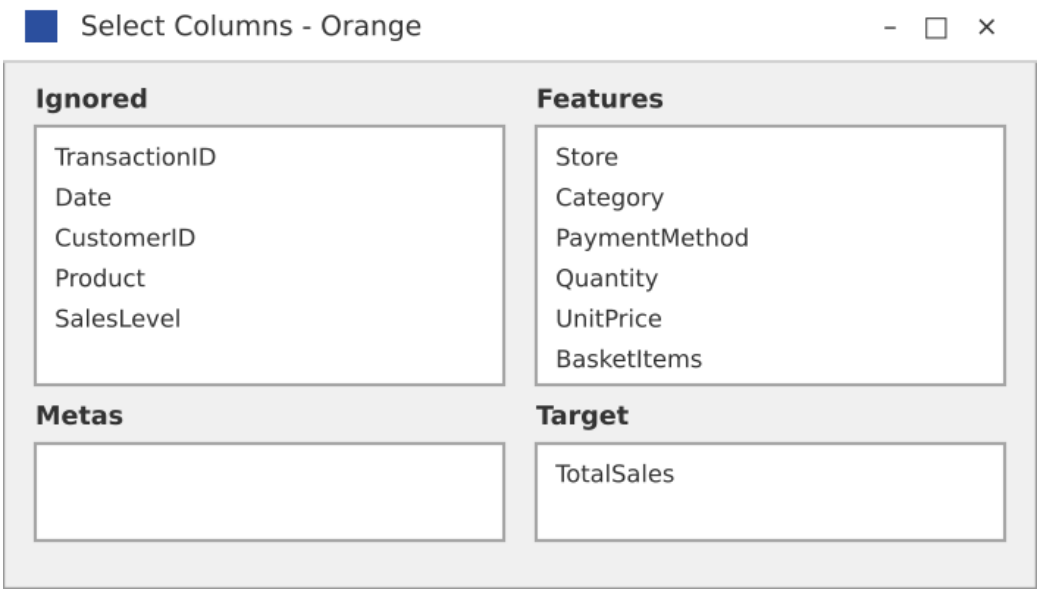


Fig 6.2 Select Columns (target = TotalSales)

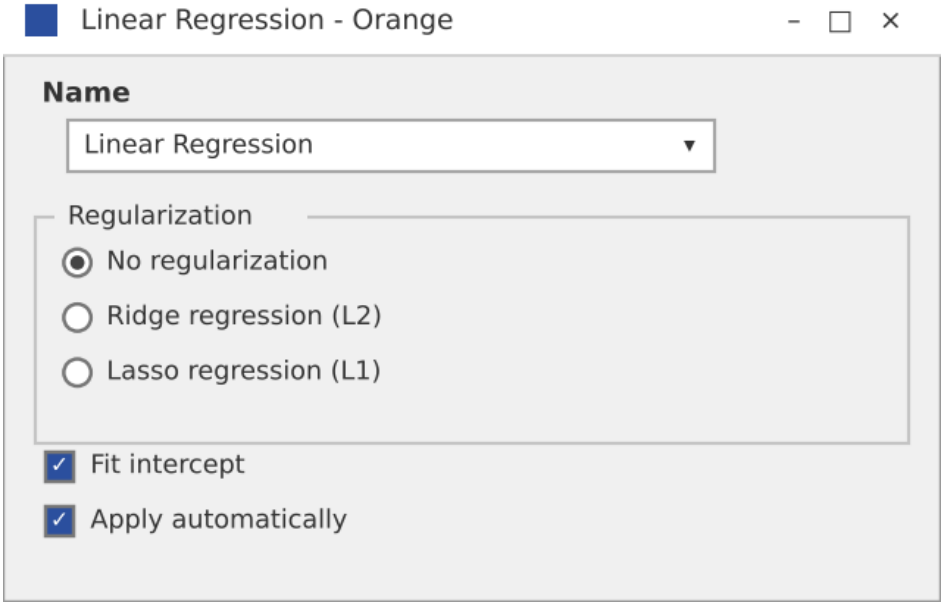


Fig 6.3 Linear Regression settings

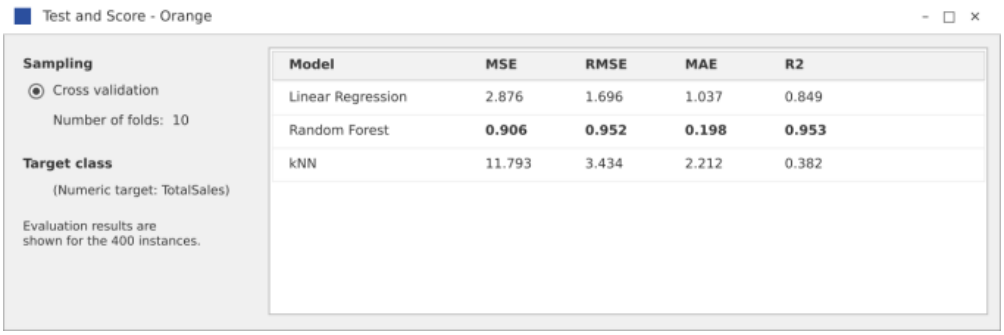


Fig 6.4 Test and Score results

Experiment 7: Outlier Analysis

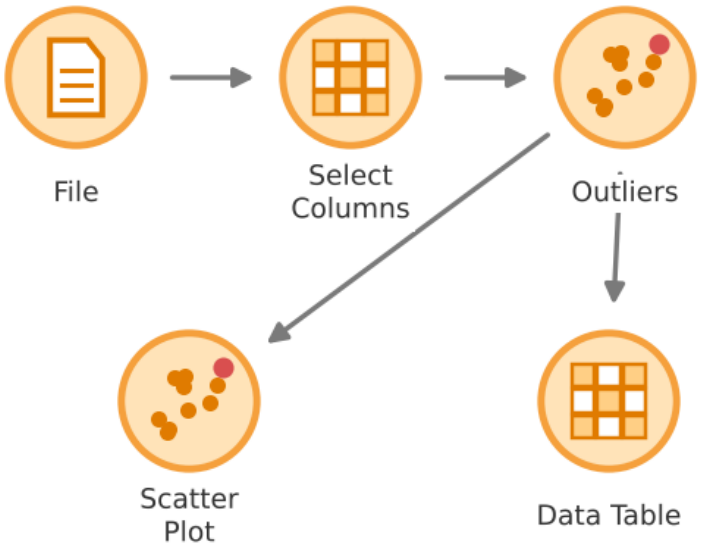


Fig 7.1 Workflow

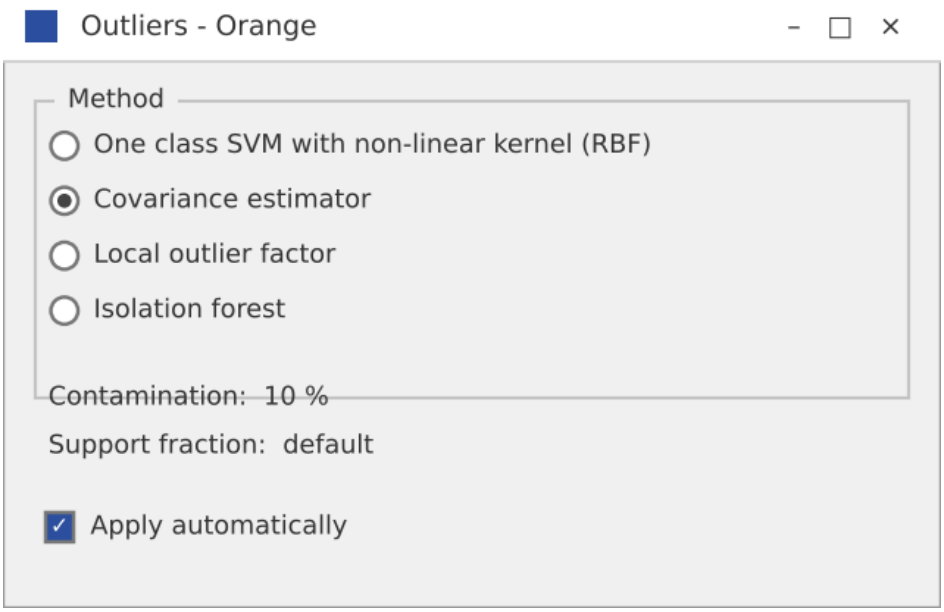


Fig 7.2 Outliers settings

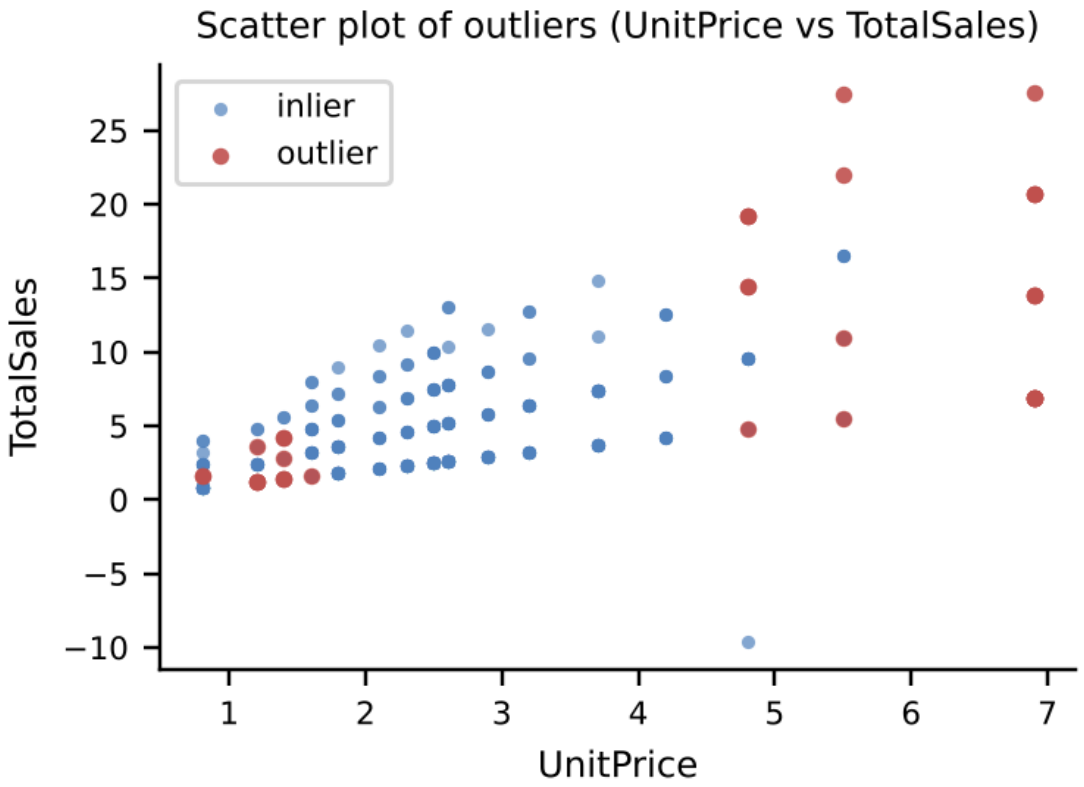


Fig 7.3 Scatter plot of outliers (UnitPrice vs TotalSales)

Data Table - Orange

	Outlier	SalesLevel	Product	Quantity	UnitPrice	TotalSales	BasketItems	Mahalanobis
253	Yes	high	Coffee	2	6.9	13.8	5	12
24	Yes	high	Coffee	1	6.9	6.9	5	11.34
327	Yes	high	Coffee	1	6.9	6.9	5	11.34
294	Yes	high	Coffee	4	6.9	27.6	3	11.22
105	Yes	high	Coffee	3	6.9	20.7	3	10.43
86	Yes	high	Coffee	3	6.9	20.7	3	10.43
383	Yes	high	Coffee	3	6.9	20.7	3	10.43
222	Yes	high	Coffee	3	6.9	20.7	4	10.17
318	Yes	high	Coffee	2	6.9	13.8	3	9.66
45	Yes	high	Coffee	2	6.9	13.8	3	9.66
49	Yes	high	Coffee	2	6.9	13.8	4	9.44
193	Yes	high	Coffee	1	6.9	6.9	3	8.92
277	Yes	high	Coffee	1	6.9	6.9	3	8.92
121	Yes	high	Coffee	1	6.9	6.9	3	8.92

Fig 7.4 Data Table of detected outliers (14 largest distances)

Experiment 8: K-Means (k = 3)

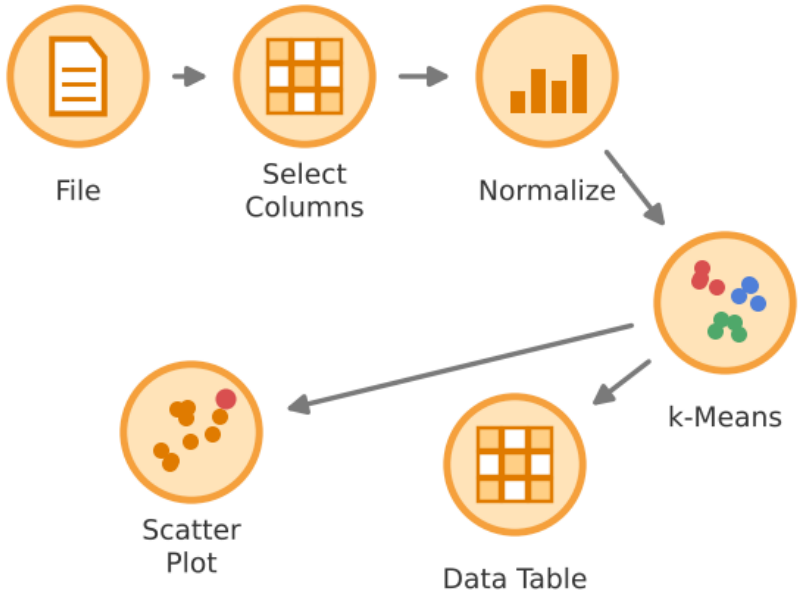


Fig 8.1 Workflow

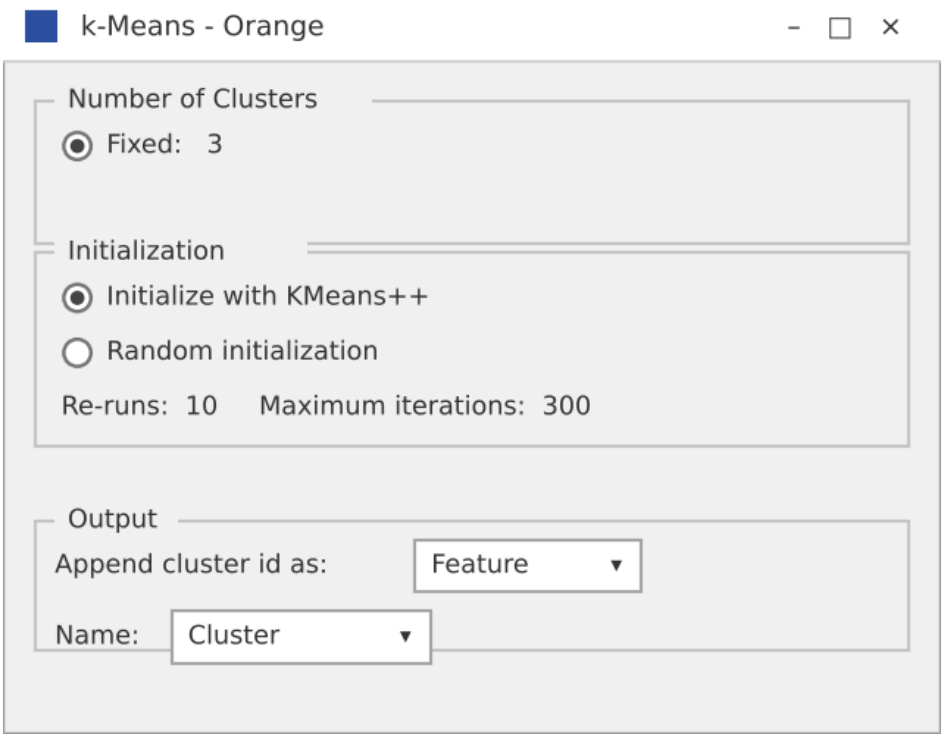


Fig 8.2 k-Means settings

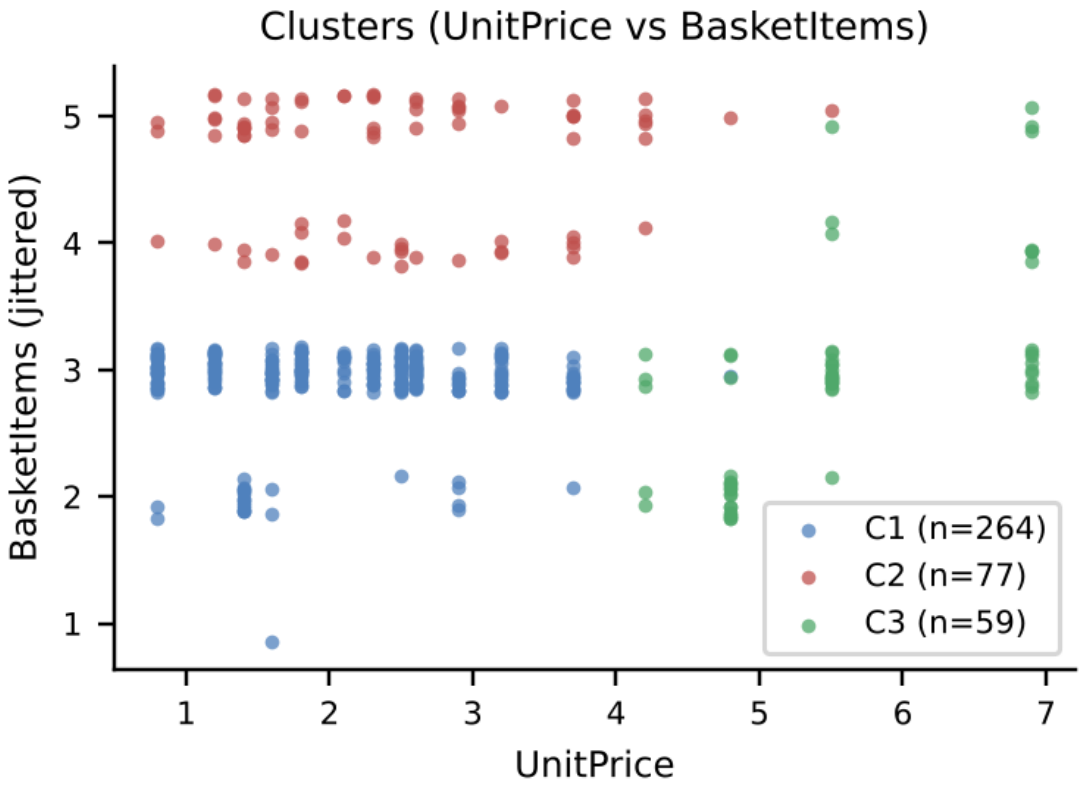


Fig 8.3 Clusters (UnitPrice vs BasketItems)

Data Table - Orange

	Cluster	SalesLevel	Category	Quantity	UnitPrice	TotalSales	BasketItems	Product
1	C1	low	Produce	2	1.2	2.4	3	Apples
2	C1	low	Produce	3	0.8	2.4	3	Bananas
3	C1	low	Beverages	1	2.6	2.6	3	Juice
4	C1	low	Bakery	1	1.8	1.8	3	Bread
5	C1	low	Dairy	1	3.2	3.2	3	Eggs
6	C1	high	Dairy	2	3.7	7.4	3	Cheese
7	C3	high	Grocery	1	5.5	5.5	3	Rice
8	C1	low	Dairy	1	2.5	2.5	3	Milk
9	C1	low	Bakery	1	1.8	1.8	3	Bread
10	C1	low	Household	1	1.4	1.4	2	Soap
11	C3	high	Household	1	4.8	4.8	2	Shampoo
12	C1	high	Snacks	3	1.6	4.8	3	Chips
13	C1	low	Snacks	1	2.3	2.3	3	Chocolate
14	C1	low	Beverages	1	2.6	2.6	3	Juice
15	C1	low	Produce	1	1.2	1.2	3	Apples
16	C1	low	Produce	1	0.8	0.8	3	Bananas

Fig 8.4 Data Table with Cluster column

Experiment 9: Simple K-Means (k = 2)

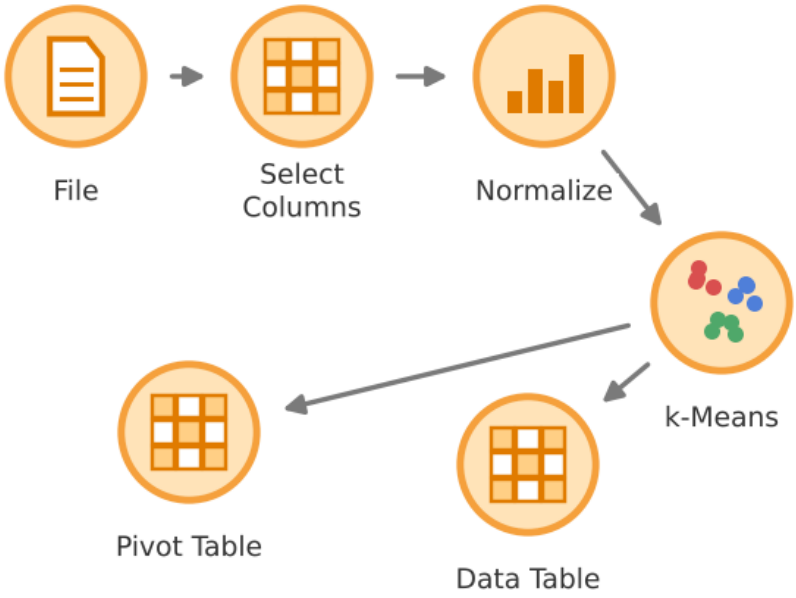


Fig 9.1 Workflow

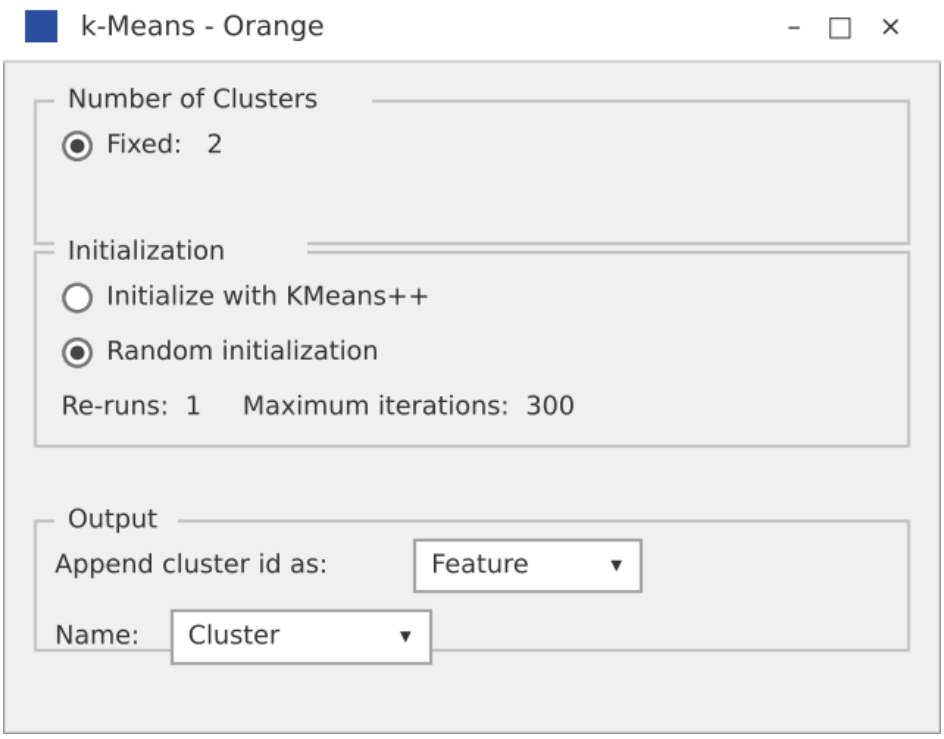


Fig 9.2 k-Means settings

Data Table (centroids) - Orange

Cluster	Size	Quantity	UnitPrice	TotalSales	BasketItems
C1	87	1.87	2.92	5.3	4.63
C2	313	1.83	2.6	4.86	2.86

Fig 9.3 Cluster centroids

Pivot Table - Orange

Rows: Cluster Columns: SalesLevel Values: Count

Cluster \ SalesLevel	high	low	Total
C1	51	36	87
C2	139	174	313
Total	190	210	400

Green = majority class of the cluster (classes-to-clusters mapping).

Fig 9.4 Classes to clusters